

DATE: 10/10/2026

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**EXNO 3: Designing a Quarter wave transformer by varying following
Parameters a) frequency b) input impedance c) characteristic impedance**

Test Case 1:

Design a quarter-wave transformer for an input impedance (Z_{in}) of 50 ohms and a load impedance (Z_L) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

AIM: To design a quarter-wave transformer for an input impedance (Z_{in}) of 50 ohms and a load impedance (Z_L) of 100 ohms and to analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.

SOFTWARE REQUIRED: SCILAB

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear;clc;  
Zr=100;s=9;d=0.1;  
Zs=50;
```

```

Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
Zof=276*log10(s/r);
Zoq=sqrt(Zr*Zof);
L=300/(f*(10^-6));
printf("length of lambda/4 transformer = %f meters\n",L);
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);

```

OUTPUT:

ZR=100 ohms Zin =50 ohms

Frequency (MHz)	Length of $\lambda/4$ transformer (meters)
100	3.000000
200	1.500000
300	1.000000
400	0.750000
500	0.600000

RESULT: A quarter-wave transformer for an input impedance (Z_{in}) of 50 ohms and a load impedance (Z_L) of 100 ohms is designed and the transformer's performance is analyzed for the varying operating frequency is varied across {100 MHz, 200 MHz, 300MHz, 500 MHz }.

Test Case 2:

For a fixed operating frequency of 2 GHz and a load impedance (Z_L) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Z_{in}) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance ($Z_{\text{transformer}} = R_o = \sqrt{Z_{in} \cdot Z_L}$) changes with Z_{in} .

AIM: To design a quarter-wave transformer a fixed operating frequency of 2 GHz and a load impedance (Z_L) of 100 ohms and evaluate its performance as the input impedance (Z_{in}) varied .To validate that the transformer's characteristic impedance ($Z_{\text{transformer}} = R_o = \sqrt{Z_{in} \cdot Z_L}$) changes with Z_{in}

SOFTWARE REQUIRED:

SCILAB

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear;
```

```
clc;
```

```
Zr=100;s=9;d=0.1;
```

```
Zs=50;
```

```
Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;
```

```
Zof=276*log10(s/r);
```

```
Zoq=sqrt(Zr*Zof);
```

```
L=300/(f*(10^-6));
```

```
printf("length of lambda/4 transformer = %f lambda\n",L);
```

```
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

OUTPUT:

ZR=100 ohms Variable :Zin

Zin	Zo
25	50.00
50	70.71
75	86.60
100	100.00

RESULT: A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (ZL) of 100 ohms and its performance is evaluated for various input impedance values. The variation of transformer's characteristic impedance with Zin is also validated..